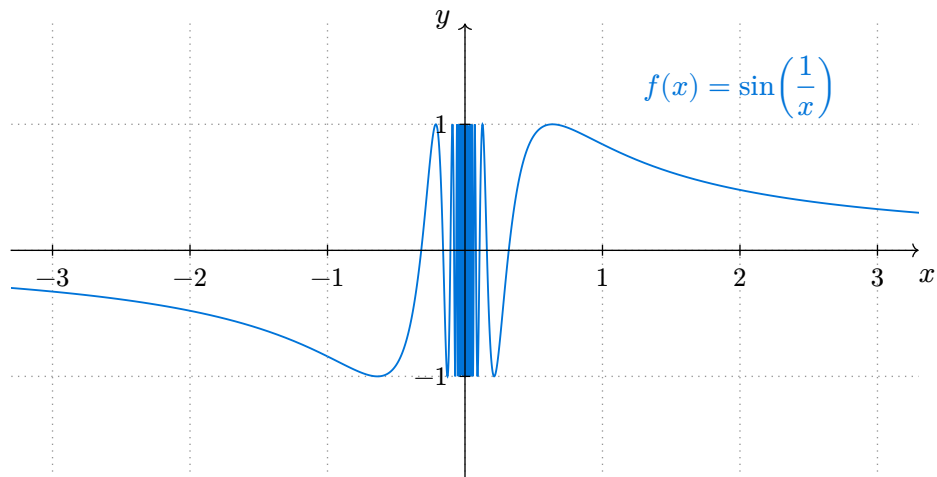


Lilaq-schoolbook

Revisit lilaq's schoolbook style.



Version 0.1.0

2026-09-10

<https://github.com/Yesteeer/typst-lilaq-schoolbook>

lilaq-schoolbook

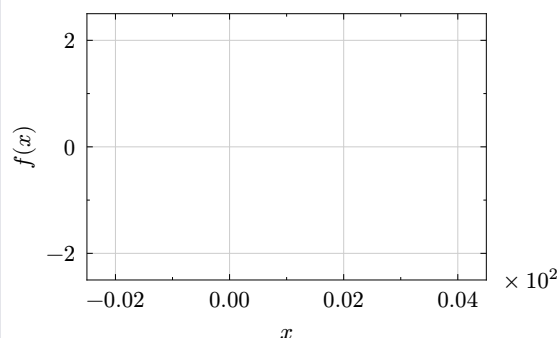
- diagram()
- plot()
- lilaq-schoolbook()

diagram

This function is used to plot functions with a fixed ratio between both axes. It is based on *lilaq*'s `diagram()` function, which arguments can be passed in the argument `sink` and will override the values computed by this package's `diagram()` function (cf. `..body`).

Example:

```
#diagram(  
  start: -2,  
  end: (4, 2),  
  offset: 0.5,  
  scale: 1.2,  
  tick-distance: 2,  
  subticks: 1,  
  add-to-xaxis: (exponent: 2),  
  labels: ($x$, $f(x)$),  
)
```



Parameters

```
diagram(  
  start: float array ,  
  end: float array ,  
  offset: float array ,  
  scale: float array ,  
  tick-distance: int array ,  
  subticks: int array ,  
  add-to-xaxis: none dictionary ,  
  add-to-yaxis: none dictionary ,  
  labels: none array ,  
  ..children: any  
) -> content
```

start float or array

Optional coordinates of the bottom left limit of the diagram A single number n is interpreted as the coordinates (n, n)

Default: $(0, 0)$

end float or array

Optional coordinates of the top right limit of the diagram A single number n is interpreted as the coordinates (n, n)

Default: $(2, 2)$

offset `float` or `array`

Optional offsets for the starting and ending coordinates.

For example, with an offset of (0.2, 0.3), the diagram starts at

$$(\text{start.x} - 0.2, \text{start.y} - 0.2)$$

and ends at

$$(\text{end.x} + 0.3, \text{start.y} + 0.3).$$

A single number n is interpreted as the coordinates (n, n) .

Default: `0.3`

scale `float` or `array`

Optional scaling along the x -axis (first coordinate) and the y -axis (second coordinate).

A single number n is interpreted as the coordinates (n, n) .

Default: `1`

tick-distance `int` or `array`

Changes the distance between consecutive ticks on the axes.

The first coordinate corresponds to the distance between ticks on the x -axis and the second one to the one on the y -axis. The default: 1 corresponds to 1cm (before scaling is applied).

A single number n is interpreted as the coordinates (n, n) .

Default: `1`

subticks `int` or `array`

Optional number of subticks between two ticks.

The first coordinate corresponds to the distance between ticks on the x -axis and the second one to the one on the y -axis. By default, there are no subticks.

A single number n is interpreted as the coordinates (n, n) .

Default: `0`

add-to-xaxis `none` or `dictionary`

Optional dictionary passed to the `xaxis` argument of `lilaq`' `diagram()` function (see [docs](#))

The dictionary does override the `tick-distance` and `subticks` arguments for the x -axis only if those are also given in the dictionary.

Default: `none`

add-to-yaxis `none` or dictionary

Optional dictionary passed to the `yaxis` argument of `lilaq`' `diagram()` function (see [docs](#))

The dictionary does override the `tick-distance` and `subticks` arguments for the x -axis only if those are also given in the dictionary.

Default: `none`

labels `none` or array

Optional axes labels.

The first coordinate corresponds to the x -axis and the second one to the y -axis.

Default: `none`

..children any

Positional arguments can be any of `lilaq`'s plot objects as well as this package's `plot()`.

Named arguments, are passed to `lilaq`'s underlying `diagram()` function.

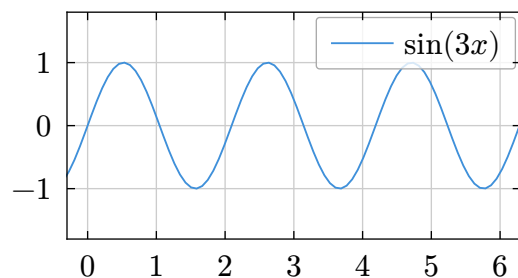
plot

This function is essentially based on `lilaq`'s `plot()` function, which arguments can be passed in the argument sink and will override the values computed by this package's `plot()` function (cf. `..args`). Its main feature is that it generates a default array of x coordinates using `lilaq`'s `linspace` function.

It needs to be passed as positional argument to a `diagram()`.

Example:

```
#diagram(  
  start: (0, -1.5),  
  end: (6, 1.5),  
  plot(  
    x => calc.sin(3*x),  
    start: -1,  
    end: 7,  
    label: $sin(3x)$  
  ),  
)
```



Parameters

```
plot(  
  y: array function ,  
  x: auto array function ,  
  start: int float ,  
  end: int float ,  
  num: int ,  
  include-end: bool ,  
  mark: none auto lilaq.mark str ,  
  ..args: any  
) -> content
```

y array or function

An array of y coordinates (see *lilaq docs*).

x auto or array or function

An array of x coordinates (see *lilaq docs*).

If set to auto, generates an array of x coordinates with *lilaq*'s `linspace` function using the start, end, num and include-end arguments below.

Default: `auto`

start int or float

Start of the range of the *lilaq*'s `linspace` function.

Default: `0`

end int or float

End of the range of the *lilaq*'s `linspace` function.

Default: `2`

num int

Number of evenly-spaced values to produce of the *lilaq*'s `linspace` function.

Default: `100`

include-end bool

Whether to include the end of the range. See *lilaq*'s `linspace` function.

Default: `true`

mark `none` or `auto` or `lilaq.mark` or `str`

The mark to use to mark data points. See *lilaq*'s `linspace` function.

Default: `none`

..args `any`

Optional arguments to be passed to *lilaq*'s `plot` function.

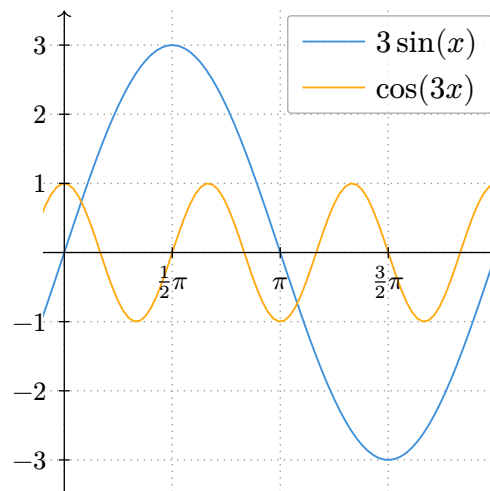
lilaq-schoolbook

A theme inspired by *lilaq*'s schoolbook theme.

Example:

```
#show: lilaq-schoolbook

#diagram(
  start: (0, -3.2),
  end: (6, 3.2),
  add-to-xaxis: (
    tick-distance: 1/2,
    locate-ticks: lq.tick-
  locate.linear.with(unit: calc.pi),
    format-ticks: lq.tick-
  format.fraction.with(suffix: $pi$),
  ),
  plot(
    x => 3*calc.sin(x),
    start: -1,
    end: 7,
    label: $3\sin(x)$,
  ),
  plot(
    x => calc.cos(3*x),
    start: -1,
    end: 7,
    label: $\cos(3x)$,
  ),
)
```



Parameters

```
lilaq-schoolbook(
  body,
  show-origin: bool
) -> content
```

show-origin `bool`

An option for showing the origin.

Default: `false`